Appendix B.
Corrected Cramér–Rao Lower Bound under the Dual-Channel AWGN Noise Model

for the SAMBPS DTaaS HIF-TF Locator (`fault_location_id`)

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Issued under WP1.6; v3 Execution Plan deliverable D-B; closes R1, R9

The IEEE Access v1 manuscript reports a Cramér–Rao analysis under a circular complex Gaussian noise model placed directly on the single-bin admittance $H_{\text{meas}} = I_{\text{bin}}/V_{\text{bin}}$. That model is *not* self-consistent with § V-B of the same manuscript, which specifies dual-channel AWGN on the time-domain voltage and current channels. H_{meas} is the ratio of two independent complex Gaussians whose density is the Marsaglia–Nadarajah–Pogány form, not Gaussian. This appendix derives two corrected bounds:

1. the **proper-complex-Gaussian-ratio CRLB** after Kuruoğlu (2018), which uses the high-SNR_V Gaussian approximation of the ratio density and includes the Geary–Hinkley validity test;
2. the **joint dual-channel FIM** in $(V(t), I(t))$ space after Nehorai & Hawkes (2000), projected onto (α, R_x).

The two are cross-checked at high SNR_V and used to overlay the empirical RMS error from the WP1.5 Monte-Carlo on the 2×4 panel `outputs/phase1_crlb_overlay/`.

Appendix B.1 Ratio-density form (Marsaglia–Nadarajah–Pogány)

For independent complex Gaussian random variables $X = \mu_X + n_X$ and $Y = \mu_Y + n_Y$ with circular complex variances σ_X^2 and σ_Y^2 per real/imag component, the ratio $Z = X/Y$ has a non-Gaussian density [1, 2] that can be written in closed form via Bessel functions but is unwieldy. In the high-SNR_Y regime, defined by the Geary–Hinkley criterion

$$\frac{|\mu_Y|}{\sigma_Y} > \tau, \quad \tau \approx 4, \tag{1}$$

a first-order Taylor expansion in n_Y/μ_Y gives

$$Z \approx \frac{\mu_X}{\mu_Y} + \frac{n_X}{\mu_Y} - \frac{\mu_X}{\mu_Y^2} n_Y, \tag{2}$$

which is itself a complex Gaussian with mean μ_X/μ_Y and *signal-dependent* per-component variance

$$\sigma_Z^2 = \frac{\sigma_X^2 + |Z|^2 \sigma_Y^2}{|\mu_Y|^2}. \tag{3}$$

The $|Z|^2 \sigma_Y^2$ term is the "ratio-shot" contribution that the v1 manuscript drops by treating σ_H^2 as constant.

Appendix B.2 Proper-ratio FIM construction

Substituting $X = I_{\text{bin}}$, $Y = V_{\text{bin}}$, $Z = H_{\text{meas}}$ in (3):

$$\sigma_H^2 \;=\; \frac{\sigma_I^2 + |H|^2\,\sigma_V^2}{|V_{\text{phase}}|^2}, \tag{4}$$

where $|V_{\text{phase}}|$ is the phasor magnitude of the ideal source at ω_0 and $\sigma_X^2 = 2\sigma_{x_t}^2/N_s$ is the per-real/imag variance of the single-bin DFT phasor of a length-N_s time-domain channel with white Gaussian noise of variance $\sigma_{x_t}^2$.

For complex observations $h(\theta) = H(\theta)$ contaminated by circular complex Gaussian noise of per-component variance σ_H^2, the Fisher information matrix on $\theta = (\alpha, R_x)$ is [3]

$$F_{kl}^{\text{proper}}(\theta) \;=\; \frac{1}{\sigma_H^2}\,\mathrm{Re}[(\partial_{\theta_k} H)^*\;\partial_{\theta_l} H]\,. \tag{5}$$

The CRLB on the variance of any unbiased estimator of θ_k is $\left(F^{\text{proper}}\right)^{-1}_{kk}$. Substituting (4):

$$F_{kl}^{\text{proper}} \;=\; \frac{|V_{\text{phase}}|^2}{\sigma_I^2 + |H|^2\sigma_V^2}\,\mathrm{Re}[(\partial_{\theta_k} H)^*\;\partial_{\theta_l} H]\,. \tag{6}$$

The partials $\partial H/\partial\alpha$ and $\partial H/\partial R_x$ are computed by the central-finite-difference fallback in `inverse_estimation/fa`
and will be swapped for the closed-form Phase-2 forms (WP2.2) when those land.

Geary–Hinkley validity. σ_H^2 in (4) is the variance of a Gaussian *approximation* of the Marsaglia density. Equation (4) is a faithful approximation only when the V channel is in the Geary–Hinkley regime (1). Outside that regime the true ratio density carries heavier tails than the Gaussian, and (6) *overestimates* the FIM (i.e. underestimates the true CRLB). The library reports a per-cell boolean `geary_hinkley_valid` flag. Under the project's $N_s = 200$ single-bin averaging the GH threshold $|V_{\text{phase}}|/\sigma_V > 4$ corresponds to time-domain $\mathrm{SNR}_V \gtrsim -8\,\mathrm{dB}$; the entire 720-case grid (which starts at 20 dB) is comfortably in the regime.

Appendix B.3 Dual-channel FIM (Nehorai–Hawkes)

The dual-channel FIM treats the raw waveform vector $\mathbf{y} = [v(t_0), \ldots, v(t_{N_s-1}),\, i(t_0), \ldots, i(t_{N_s-1})]^T$ as the observation, with $v(t_n) = V_{\text{phase}}\cos(\omega_0 t_n) + n_v(n)$ and $i(t_n) = \mathrm{Re}\{H(\theta)V_{\text{phase}}e^{j\omega_0 t_n}\} + n_i(n)$. Under the project's ideal-source assumption $\partial v/\partial\theta \equiv 0$, so the V channel contributes zero to the FIM; only the I channel carries information about θ. Defining the Jacobian $[J_i]_{n,k} = \partial i_{\text{clean}}(t_n)/\partial\theta_k$,

$$F_{kl}^{\text{dual}}(\theta) \;=\; \frac{1}{\sigma_{i_t}^2}\sum_{n=0}^{N_s-1}\frac{\partial i_{\text{clean}}(t_n)}{\partial\theta_k}\,\frac{\partial i_{\text{clean}}(t_n)}{\partial\theta_l}. \tag{7}$$

A one-cycle window with N_s samples evaluates the inner sum analytically (cosine–cosine and sine–sine sums each contribute $N_s/2$, cross terms cancel):

$$F_{kl}^{\text{dual}} \;=\; \frac{N_s}{2}\,\frac{V_{\text{phase}}^2}{\sigma_{i_t}^2}\,\mathrm{Re}[(\partial_{\theta_k} H)^*\;\partial_{\theta_l} H]\,. \tag{8}$$

Appendix B.4 Cross-check: proper-ratio vs dual-channel

The same Fisher kernel $\mathrm{Re}[(\partial_k H)^*\,\partial_l H]$ appears in both (6) and (8); the prefactors differ. Substituting $\sigma_I^2 = 2\sigma_{i_t}^2/N_s$ and $\sigma_V^2 = 2\sigma_{v_t}^2/N_s$:

$$\frac{F^{\text{proper}}}{F^{\text{dual}}} \;=\; \frac{\sigma_I^2}{\sigma_I^2 + |H|^2\sigma_V^2}. \tag{9}$$

Two limiting cases:

- $\sigma_V \to 0$ (V perfectly known): the two FIMs are *exactly equal*. This is the **WP1.6 brief acceptance regime** ("agree to within 5 % at $\mathrm{SNR}_I \geq 40\,\mathrm{dB}$ in the Geary–Hinkley regime") verified by `tests/test_crlb_consistency.py::test_crlb_proper_eq_dual_when_V_noiseless`.

- $\sigma_V = \sigma_I/|H|$ (V and I shot noise balanced): $F^{\mathrm{proper}} = F^{\mathrm{dual}}/2$, i.e. a factor of two loss in information by accessing only the ratio instead of the raw waveforms. In RMSE terms, $\mathrm{RMSE}^{\mathrm{proper}} = \sqrt{2}\,\mathrm{RMSE}^{\mathrm{dual}}$. Verified by `test_crlb_factor_sqrt2_when_snrV_eq_snrI`.

Direction of the inequality. Note $F^{\mathrm{proper}} \leq F^{\mathrm{dual}}$ always, with equality iff $\sigma_V = 0$. The proper-ratio CRLB is therefore $\geq$ the dual-channel CRLB (it represents *less* information about θ because the H-ratio observation discards the absolute magnitudes that the raw V and I waveforms preserve). This is opposite to the v1 "Gaussian-on-H" linearisation, which artificially *tightens* the bound by ignoring the $|H|^2\sigma_V^2$ term in (4).

Appendix B.5 Headline empirical-vs-CRLB result

The WP1.5 Monte-Carlo (100 trials per cell on the 720-case grid) provides empirical per-cell $\mathrm{RMSE}(\hat{\alpha})$ values that the bounds (6) and (8) are plotted against in the 2×4 overlay panel `outputs/phase1_crlb_overlay/`. The four headline findings this Appendix supports:

1. The proper-ratio bound is the *correct* bound on any unbiased estimator that operates on H_{meas}. It supersedes the v1 "Gaussian-on-H" linearisation, which is invalid outside the asymptotic $|I| \gg \sigma_I$ regime.

2. The dual-channel bound is the ground truth in (V, I) waveform space; it is *tighter* (lower) than the proper-ratio bound by the factor in (9).

3. The two bounds agree to within 5 % (in fact exactly) at $\sigma_V \to 0$, certified by `test_crlb_proper_eq_dual_whe`

4. The Geary–Hinkley validity flag holds across the full 720-case grid (time-domain SNR_V down to $-8\,\mathrm{dB}$), so (4) is a faithful approximation throughout. Outside this regime the proper- ratio FIM *overestimates* the true Marsaglia FIM.

Appendix B.6 Multi-port projection (WP3.6)

The single-port derivation in Sect. B.1–B.4 treats the IED's observation as a single complex bin $H_{\mathrm{meas}} = I_{\mathrm{bin}}/V_{\mathrm{bin}}$. In the three-phase case the IED instead observes the sending-end admittance *matrix* $Y_{\mathrm{send}} \in \mathbb{C}^{3\times 3}$ relating $I_{\mathrm{abc}} = Y_{\mathrm{send}}\,V_{\mathrm{abc}}$ at the fundamental. This section derives the multi-port FIM and certifies its consistency with the single-port limit.

Per-entry proper-ratio variance. Each entry Y_{pq} is the ratio of two complex Gaussians (I_p phasor over V_q phasor) under the WP1.6 Geary–Hinkley regime, so the per-entry approximate complex-Gaussian variance is

$$\sigma^2_{Y_{pq}} = \frac{\sigma^2_{I_p} + |Y_{pq}|^2\sigma^2_{V_q}}{|V^{(q)}_{\mathrm{phase}}|^2} \quad p, q \in \{a, b, c\}, \tag{10}$$

matching (4) per entry.

Multi-port proper-ratio FIM. The three-phase observation is the stacked vector $y \in \mathbb{R}^{2\,|\mathcal{S}|}$ of Re/Im parts of the chosen entry set $\mathcal{S} \subseteq \{(p, q) : p, q \in \{a, b, c\}\}$. Three observation policies are exposed in the public API:

- $\mathcal{S}_{\mathrm{full}}$ – all 9 entries (18 real),

- $\mathcal{S}_{\text{upper}}$ – 6 upper-triangle entries (12 real),
- $\mathcal{S}_{\text{diagonal}}$ – 3 diagonal entries (Y_{aa}, Y_{bb}, Y_{cc}, 6 real).

Stacking the per-entry contributions gives the 2×2 FIM

$$F_{kl}^{\text{proper}} = \sum_{(p,q)\in\mathcal{S}} \frac{1}{\sigma_{Y_{pq}}^2} \operatorname{Re}\Big[\big(\partial_{\theta_k} Y_{pq}\big)^* \partial_{\theta_l} Y_{pq}\Big], \tag{11}$$

with $\theta = (\alpha, R_x)$ and $\mathrm{CRLB}(\theta_k) = (F^{-1})_{kk}$.

Multi-port dual-channel FIM. Following the WP1.6 Nehorai–Hawkes derivation in waveform space, each per-entry contribution to the FIM becomes

$$F_{kl}^{\text{dual}} = \sum_{(p,q)\in\mathcal{S}} \frac{N_s\,|V_{\text{phase}}|^2}{2\,\sigma_{I_{pq,t}}^2} \operatorname{Re}\Big[\big(\partial_{\theta_k} Y_{pq}\big)^* \partial_{\theta_l} Y_{pq}\Big], \tag{12}$$

with $\sigma_{I_{pq,t}}^2$ the time-domain variance of the per-(p,q) current and $N_s = 200$ the cycle window length. The two FIMs share the same kernel $\operatorname{Re}[(\partial Y_{pq})^* \partial Y_{pq}]$ summed over $\mathcal{S}$; only the per-entry weights differ.

Cross-check: consistency at $\sigma_V \to 0$. At $\sigma_V \to 0$ the proper-ratio per-entry weight in (11) reduces to $1/\sigma_{Y_{pq}}^2 = |V_{\text{phase}}|^2/\sigma_{I_{pq,b}}^2$ where $\sigma_{I_{pq,b}}^2 = (2/N_s)\sigma_{I_{pq,t}}^2$ is the single-bin DFT variance. Substituting yields exactly the prefactor in (12), so $F^{\text{proper}} = F^{\text{dual}}$ entry-by-entry under V-noiseless balanced 3-phase operation. The WP3.6 numerical check on the IEEE 34 720-grid sub-sample reports $\max_{\text{cell}} |\, r - 1\,| \le 7{\times}10^{-16}$ where $r = \text{rmse}_\alpha^{\text{proper}}/\text{rmse}_\alpha^{\text{dual}}$ (300/300 cells within 5 %; in fact at machine precision). Cross-checked in `tests/test_fim_multiport.py`.

Information accumulation. The multi-port FIM is a sum of $|\mathcal{S}|$ per-entry contributions; under balanced 3-phase operation each entry's kernel is approximately equal to the single-port kernel scaled by a constant. In the asymptotic-MLE limit the multi-port CRLB is therefore tighter than the single-port bound by roughly $\sqrt{|\mathcal{S}|}$ – a $\sqrt{9} = 3\times$ improvement for the full observation in the no-load limit. Empirically on the IEEE 34 sub-sample (load-dominated Y_{send}) the per- cell improvement is smaller because the load admittance dilutes the per-entry parameter sensitivity $\partial Y_{pq}/\partial\theta$; the WP3.6 commit ships overlay PNGs in `outputs/phase3_crlb_multiport_overlay/` that visualise this load-dilution effect. The closure of the high-R_x near-source corner – where the load dilution dominates the noise floor – closes structurally at the WP3.5 Taylor–Fourier multi-bin extension already landed.

End of Appendix B.

References

[1] G. Marsaglia, "Ratios of normal variables and ratios of sums of uniform variables," *Journal of the American Statistical Association*, vol. 60, no. 309, pp. 193–204, 1965.

[2] S. Nadarajah and T. K. Pogány, "On the distribution of the product of correlated normal random variables: Closed-form expression and computational aspects," *Comptes Rendus Mathématique*, vol. 354, pp. 201–204, 2018, cited in support of the Marsaglia-Nadarajah-Pogány ratio density form referenced in § VIII.

[3] S. M. Kay, *Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory*. Prentice Hall, 1993.